

RESERVOIR RESPONSE STUDY

Numerical Methods · Lab 02 · Alimane · BES6-M

FITTED PARAMETERS · TWO LOGISTIC COMPONENTS

Parameter	Estimate	SE	t	p-value
c · baseline	14.34926	0.00968	1483.06	0
a · rise amplitude	8.91606	0.75314	11.84	1.67e-26
k · rise rate	0.51173	0.02128	24.05	3.6e-70
t · rise midpoint	31.51980	0.09055	348.08	0
a · recession amplitude	-3.72130	0.75940	-4.90	1.62e-06
k · recession rate	0.28312	0.01375	20.59	4.79e-58
t · recession midpoint	37.54602	1.05748	35.51	7.14e-106

MODEL AND INTEGRATION SUMMARY

SSE	R ²
1.8017	0.999114
STANDARD ERROR	DEGREES OF FREEDOM
0.0801 m	281
FITTED AREA	QUAD ABSOLUTE ERROR
1260.9708 m·h	5.08e-07
RAW TRAPEZOID CHECK	PEAK FITTED DH/DT
1260.9963 m·h	1.0060 m/h at 31.32 h

RESIDUAL READING

The residuals have 17 sign runs across 288 readings. Their maximum absolute value is 0.221 m, which is 22.1 times the logger resolution. The high R² is useful, but sustained signs show that the simple model does not reproduce every local feature.